
Climate Canary

Raumüberwachung mit Fokus auf
Datenschutz, Zuverlässigkeit und
Benutzerfreundlichkeit

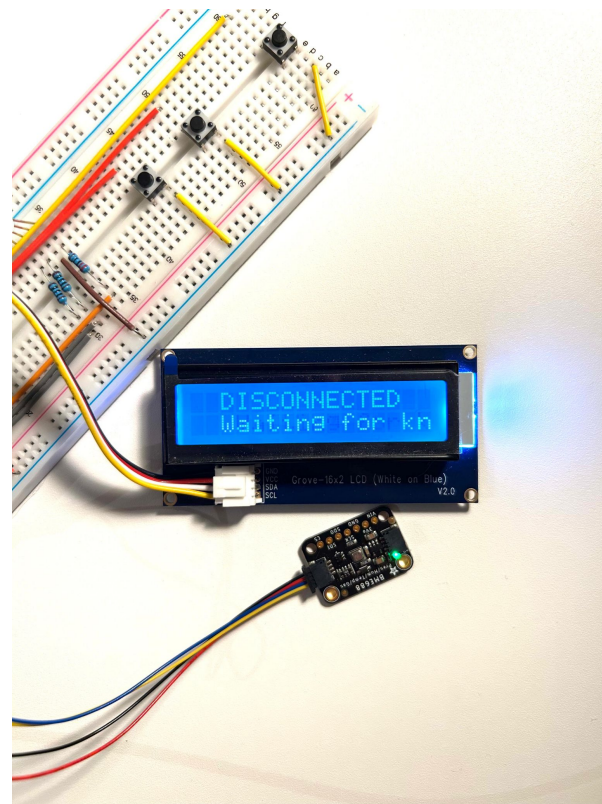
PS Software Engineering SS/2026

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Zuverlässigkeit und Ausfallsicherheit

- Benutzerfreundliches Setup
- LED Zustände mit Farben & Blinkmustern
- Live-Status direkt am Arduino Display
- Detailliertes Logging am Raspberry PI
- Kein Datenverlust bei Verbindungsproblemen:
 - Bis zu 1000 Messwerte lokal gespeichert
 - Automatische Wiederübertragung
 - 48 Retry-Mechanismus zwischen RPI und Backend



```
2026-05-23T23:05:47 INFO [SYS] adding station 7C:DE:CE:44:CC:B1
2026-05-23T23:05:47 INFO [BLE:7C:DE:CE:44:CC:B1] connecting (pi_id=16, sensor_station_id=15, name='sensor-station-1')...
2026-05-23T23:05:49 INFO [BLE:7C:DE:CE:44:CC:B1] connected.
2026-05-23T23:05:49 INFO [BLE:7C:DE:CE:44:CC:B1] authenticated (pi_id=16, room='Common Area 2')
2026-05-23T23:05:49 INFO [BLE:7C:DE:CE:44:CC:B1] reading cached sensor data...
2026-05-23T23:05:49 INFO [BLE:7C:DE:CE:44:CC:B1] cached read done: 2 packet(s) collected, timestamps deferred
2026-05-23T23:05:50 INFO [BLE:7C:DE:CE:44:CC:B1] subscribed to notifications
2026-05-23T23:06:39 INFO [BLE:7C:DE:CE:44:CC:B1] timestamp anchor set at pi=1779570399.313, arduino_ms=55007
2026-05-23T23:06:39 INFO [BLE:7C:DE:CE:44:CC:B1] cached [0] 2026-05-23T23:05:47.305: 21.4°C 45.0% 95846.2hPa 50.0%
2026-05-23T23:06:39 INFO [BLE:7C:DE:CE:44:CC:B1] cached [0] queued for upload
```



Rollenbasierte Zugriffslogik

- **EMPLOYEE** → eigenes Büro & common areas
- **DEP LEAD** → Department Overview
- **MANAGEMENT** → anonymisierte Aggregationen
- **BUILDING ADMIN** → Vollzugriff

Room-Fair Aggregation

- Räume werden zuerst einzeln aggregiert
- Danach erst Gesamttaggregation
- Verhindert unfairen Einfluss einzelner Räume

Bucket-Logik – resolveBucketSize()

≤ 7 Tage → “raw”

Keine Aggregation, jeder Messwert wird einzeln mit timestamps zurückgegeben

≤ 90 Tage → “1h”

Messwerte werden stündlich gemittelt

≤ 1 Jahr → 6h

Messwerte werden in 6h Blöcken gemittelt

default → 1d

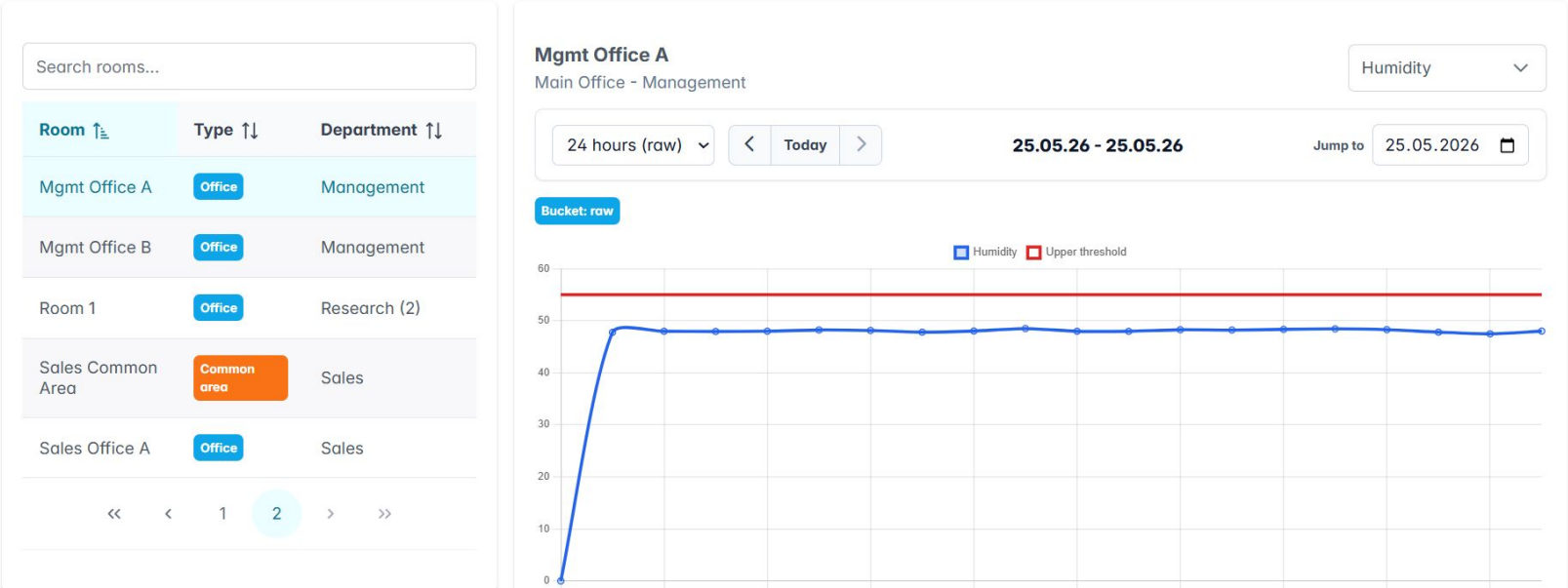
Messwerte werden in 6h Blöcken gemittelt

Department Lead: ≤ 30 Tage → 1d, default 1w
(forceReduced())

Frontend und User Experience

Room Climate Data

Building administration view for persisted measurements, trends, and threshold violations.



Frontend und User Experience

[Dashboard](#)

[Absences](#)

[Overview](#)

[\[-> Logout\]](#)

**@iris**
Iris Klein

Welcome, Iris

 Department climate changes and warnings

**5**
Departments

**6**
Warnings

**5**
Departments affected

**Engineering**

**2 warnings**

Week and month compared with previous equal period

**Humidity: 1**

**Air Quality: 1**

STATISTIC	LAST WEEK	LAST MONTH
Humidity	 Improved	— Unchanged
Temperature	— Unchanged	— Unchanged
Pressure	 Improved	— Unchanged
Air Quality	 Worsened	 Worsened

**HR**

**1 warning**

Week and month compared with previous equal period

**Humidity: 1**

STATISTIC	LAST WEEK	LAST MONTH
Humidity	 No data	 No data
Temperature	 No data	 No data
Pressure	 No data	 No data
Air Quality	 No data	 No data

**Management**

**1 warning**

Week and month compared with previous equal period

**Temperature: 1**

STATISTIC	LAST WEEK	LAST MONTH
Humidity	 No data	 No data
Temperature	 No data	 No data
Pressure	 No data	 No data
Air Quality	 No data	 No data

Frontend und User Experience



Filter by Employee

All employees

Absence Types

Holiday

Sick Leave

Parental Leave

Other

Month

Week

Agenda

<

Today

>

May 2026

Jump to

25.05.2026

Holiday

Sick Leave

Parental Leave

Other

Mon	Tue	Wed	Thu	Fri	Sat	Sun
27	28	29	30	01	02	03
04	05	06	07	08	09	10
Ben Schmidt — Holiday						
11	12	13	14	15	16	17

Team Absences

Engineering

Manage Absences

Absence Overview

Employee

All employees

Status

All statuses

From

Start

To

End

4 absences

1 pending approval

Ben Schmidt

Holiday

PERIOD

05/05/2026 – 09/05/2026

Planned

Approve

Reject

Ben Schmidt

Sick Leave

PERIOD

03/02/2026 – 04/02/2026

Approved

Clara Huber

Holiday

PERIOD

10/02/2026 – 14/02/2026

Approved

Frontend und User Experience

 Dashboard  Absences  Building 2 

 Logout  @eva
Eva Kowalski

 Thresholds  Climate Hints



 Eng Office A 4 thresholds

HUMIDITY 1 threshold

Type ↑	Limit Value ↑↓	Enabled	Actions
LOWER	65		 

TEMPERATURE 2 thresholds

Type ↑	Limit Value ↑↓	Enabled	Actions
LOWER	18		 
UPPER	27		 



Frontend und User Experience

Scan for Arduino Devices

The Raspberry Pi will scan for nearby Arduino devices in setup mode (takes ~10 seconds). Select a found device to register it as a sensor station.

Start Scan

BLE MAC Address	Action
7C:DE:CE:44:CC:B1	Select

Close

Sensor Stations — rpi-test25

Download conf.yaml

Scan for Arduinos

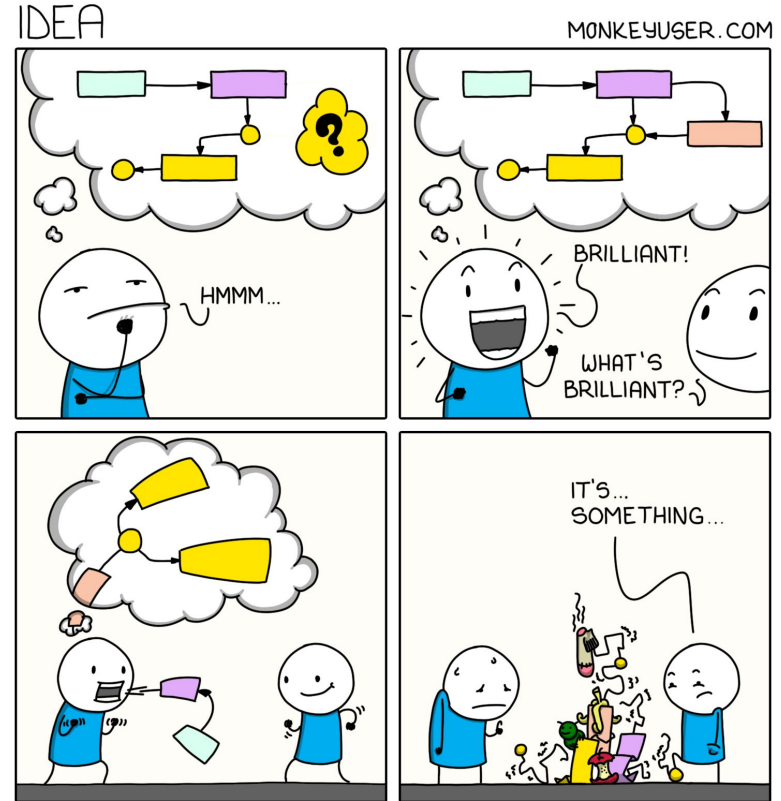
ID	Name	Status
17	7C:DE:CE:44:CC:B1	Needs Setup

Interval (s) **Status** **Actions**

58	AVAILABLE	
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Entwicklung & Teamarbeit

- Agiles Projektmanagement
- Modularer Systemaufbau
- Inkrementeller Entwicklungsprozess
- Kontinuierliche Lauffähigkeit des Systems
- Selbstorganisation und Eigeninitiative im Team
- Echtzeit-Kommunikation über Instant Messaging
- Gemeinsame Ausarbeitung von Konzept und Systemdesign





Live Demo

